

ALEX BAKER

Undergraduate Physics Student · Computational Astrophysics & Scientific Machine Learning

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PROFILE

Undergraduate physics student at Université Laval working at the intersection of astrophysics, machine learning, and scientific computing. My work focuses on building interpretable and physically grounded pipelines for astronomical survey data, with current projects in stellar spectroscopy, galaxy morphology, and Bayesian cosmological inference. Preparing for graduate studies in computational astrophysics, with a particular interest in large astronomical surveys and physics-driven machine learning.

EDUCATION

B.Sc. Physics — Université Laval

Sept. 2023 – present

Département de physique, de génie physique et d'optique · Québec, QC, Canada

- Focus on astrophysics, scientific computing, and machine learning applications in physics.
- Course research project on dimensionality reduction and interpretability for stellar spectral analysis using LAMOST DR5 × Gaia DR3 data; presented at the PHY-3500 departmental symposium, Musée de la Civilisation de Québec, Winter 2026. Supervised by Prof. Antoine Allard.

RESEARCH PROJECTS

AstroSpectro — Interpretable ML Pipeline for Stellar Spectroscopy

Apr. 2025 – present

github.com/PhD-Brown/AstroSpectro · phd-brown.github.io/AstroSpectro

- Developed an end-to-end interpretable machine-learning pipeline for stellar spectral analysis using 43,000+ LAMOST DR5 spectra cross-matched with Gaia DR3.
- Engineered 183 physics-based spectroscopic descriptors, including Balmer-line features, metallic-line indicators, molecular bands, continuum indices, and line-profile measurements.
- Trained and evaluated XGBoost-based classifiers; explicitly tested the effect of removing instrumental, positional, and potentially leaky metadata (ROC-AUC 0.964, 87% balanced accuracy).
- Applied PCA, UMAP, t-SNE, autoencoders, clustering, and SHAP interpretability to study the physical structure of the learned spectral feature space.
- Current SHAP results suggest that metallicity-sensitive features (Ca II H&K; and Mg b) play a major role in the learned classification structure alongside classical Balmer temperature indicators.
- Manuscript in preparation; target venue under consideration. 85+ experiment runs tracked with Weights & Biases.

AstroVision — Deep Learning for Galaxy Morphological Classification

Apr. 2025 – present

github.com/PhD-Brown/AstroVision

- Developed a deep-learning pipeline for galaxy morphological classification on Galaxy10 DECaLS (17,736 images, 10 classes), combining DINOv2 visual features, non-parametric morphometrics (CAS, Gini, M20), and SDSS photometry.
- Benchmarked CNN, EfficientNet-B0, DINOv2 probing, fine-tuning, and late-fusion multimodal strategies; best model achieves 86.3% balanced accuracy.
- Integrated Monte Carlo Dropout uncertainty quantification and conformal prediction to identify ambiguous morphology classifications.
- Prepared an A&A-style; internal manuscript draft and a reproducible codebase.

ξ Dark Energy · BAO MCMC — Bayesian Cosmological Inference

Jan. 2025 – present

github.com/PhD-Brown/xi-dark-energy-bao-mcmc

- Implemented a reproducible Bayesian MCMC pipeline comparing Λ CDM, CPL, and exploratory oscillating dark-energy parameterisations using Pantheon+SH0ES supernovae and DESI DR2 BAO measurements.
- Built likelihood functions for supernova and BAO distance observables, posterior diagnostics, corner plots, and AIC/BIC model comparison.
- Released as v0.1.0-exploratory with explicit documentation of modelling limitations, including fixed sound horizon assumptions, prior-boundary effects, and self-consistency issues in the oscillating model.

INTELLECTUAL PROPERTY

Co-Inventor — MOSS System (Provisional Patent)

May 2025 – present

Modular Orchestrated Storage System — persistent memory architecture for large language models

- Co-inventor on US Provisional Patent Application No. 63/882,562.
- Contributed to the design of the orchestration API and developed Python prototypes for persistent-memory workflows.
- Project currently under technology-transfer and valorisation discussions with Axelys and BLUM, Université Laval.

TEACHING EXPERIENCE

Teaching Assistant — Classical Mechanics (PHY-1005)

Winter 2025

Département de physique, de génie physique et d'optique, Université Laval

- Academic support for undergraduate students in mechanics and special relativity.
- Explanation of complex physical concepts and correction of mathematical problems.

TECHNICAL SKILLS

Programming	Python · NumPy · SciPy · pandas · Astropy · Matplotlib · Plotly · LaTeX · Bash/Linux
Machine Learning	scikit-learn · XGBoost · PyTorch · SHAP · UMAP · HDBSCAN · DINOv2 · EfficientNet · MC Dropout · Weights & Biases
Astronomy & Cosmology	FITS processing · LAMOST DR5 · Gaia DR3 · SDSS · Pantheon+SH0ES · DESI BAO · emcee · Vizier · CDS XMatch
Tools	Git · GitHub · GitHub Pages · Docusaurus · Jupyter · VS Code · CUDA · joblib

MANUSCRIPTS, REPORTS & RESEARCH OUTPUTS

[1] Baker, A. (2026, in preparation). Interpretable Machine Learning for Stellar Spectral Classification with LAMOST DR5 and Gaia DR3. *Manuscript in preparation; target venue under consideration.*

[2] Baker, A., Jean, J., & Caillat, N. (2026). Réduction de dimension et interprétabilité appliquées aux spectres stellaires LAMOST DR5. *PHY-3500 Course Report & Symposium Contribution*, Université Laval. Supervised by Prof. Antoine Allard.

[3] Baker, A. (2025). ξ Dark Energy · BAO MCMC: Exploratory Bayesian Inference for Dark-Energy Models. *Research software release, v0.1.0-exploratory*. github.com/PhD-Brown/xi-dark-energy-bao-mcmc

LANGUAGES

French	Native	English	Fluent — scientific writing and communication
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